University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 2, 2025

Answer all questions.
Marks per question are shown in brackets.

Time: 45 minutes (+5 minutes reading time) **Marks: 35**

Section 1: Theory of Algorithms [20]

Question 1. **[4]**

Select all options that apply in the following questions.

a) if $T(n) = n^2 + (\log n)^2$, then T(n) is in

i. $O(n^3)$
ii. $\Theta(n^2)$
iii. both i. and ii.
iv. none of the above

b) Is the following statement True or False?

"We can use the Master Theorem to solve a recurrence of the form $T(n) = aT(n-b) + f(n)$, where f(n) is in $\Theta(n^d)$"

c) Given $T(n) = T(n/3) + 3$, the solution to the recurrence is in

i. $\Theta(n^2)$
ii. Θ (log n)
iii. Θ (n)
iv. Θ (1)

d) Solving the Travelling Salesman (TSP) problem, between *n* cities, by brute force enumeration (exhaustive search) is in

i. $\Theta(2^n)$
ii. $\Theta(n^n)$
iii. Θ (n!)
iv. none of the above

Question 2. **[3]**

You are given an unsorted array of *n* elements and asked to find the kth smallest element.

a) Explain briefly how you could do this *without* first sorting the array. [2]

b) Why will this approach not always be better than presorting? [1]

Question 3. **[2]**

Why does *interpolation search* not always reduce the search domain by a constant factor like binary search?

Question 4. **[4]**

You are given an array of elements [s_1, ..., s_n] and asked to write an algorithm to sum all the elements. Your classmate claims you can improve on brute force summation (n-1 adds) by using a simple *divide and conquer* algorithm to do the summation in "log n operations"! Show that this is not true by defining and solving the recurrence relation for the number of *additions* A(n) required to compute the full sum using the obvious divide-and-conquer approach for n elements. After you have defined the full recurrence relation, you can assume that $n = 2^k$ to solve it – you may use the Master Theorem. NOTE: this is a *serial* solution, so no parallel sums.

Question 5. **[4]**

Use Horner's algorithm for exponentiation to compute the number of multiplies needed to compute a^{13}. **Show all your work**.

Question 6. **[3]**

You are given the following solution table for the knapsack problem of capacity W = 6 and n=5 items.

		capacity j						
	i	0	1	2	3	4	5	6
	0	0	0	0	0	0	0	0
$w_1 = 3, v_1 = 25$	1	0	0	0	25	25	25	25
$w_2 = 2, v_2 = 20$	2	0	0	20	25	25	45	45
$w_3 = 1, v_3 = 15$	3	0	15	20	35	40	45	60
$w_4 = 4, v_4 = 40$	4	0	15	20	35	40	55	60
$w_5 = 5, v_5 = 50$	5	0	15	20	35	40	55	65

Use this information determine the most valuable items that can fit into a knapsack of capacity W'= **5**. Show your reasoning – based on table values - used to determine the solution set – guess will not be accepted.

Section 2: Theory of Computation [15]

Question 7. **[7]**

a) Using B to represent a blank cell, **draw a transition diagram for a Turing Machine** with alphabet {B,1} that does natural number addition for unary numbers, and stops with

the tape head at the cell immediately right of the total. A number N is represented by a string of N+1 consecutive 1s. The tape head will start on a blank cell, followed by the two numbers, with a single blank cell between them. Example input for 3 + 1:

```
B1111B11BB- - -
↑
```

which must generate output:

```
B11111BBBB- - -
        ↑
```

[5]

b) **How many rows** would your Turing Machine above have in its State Table? [1]

c) Give an example of **any one row** in your Turing Machine's State Table. [1]

Question 8. **[3]**

Give a regular expression (**regex**) that accepts the language AskLess:

AskLess = { w | w comprises 2 binary numbers separated by either "<" or "<=" }

Example strings in this language: 10101<=11011 or 11011<=10101 etc. – i.e. a language for stating that kind of Boolean expression.

Question 9. **[1]**

We know there are languages no Turing Machine can recognize because (A) the set of all languages is *uncountably* infinite; and (B) the set of all Turing Machines is *countably* infinite. **In one sentence**, **explain** briefly how we know (B) is true.

Question 10. **[4]**

Which of the following statements are **false**? **Give the letter(s) of the false** statement(s), with a brief reason for your answer. **Negative marking employed**: right minus wrong! Minimum mark zero.

a) If the input of a problem is of size N, then the lower bound of that problem must be at least Ω(N).
b) Using the adversary argument in a "Guess the 2-digit number" game, one can keep changing the correct answer after each guess. For example, start with 48, then make it 62, then 55, etc.
c) Any language that is decidable is also acceptable.
d) If problem A has a lower bound of N log N, and we can show that problem B is reducible to problem A in linear time, then problem B also has a lower bound of N log N.

ANSWER:

(a) is false, e.g. binary search

(b) is false, as 2nd value must be <48, because fewer 2-digit numbers below 48 than above it
(d) is false, we'd need to show A is reducible to B, i.e. the other way round is correct

Appendix A – Master Theorem

If we have a recurrence of this form: $T(n) = aT(n/b) + f(n)$, $f(n) \in \Theta(n^d)$ and $T(1) = c$ then

$T(n) \in \Theta(n^d)$ if $a < b^d$

$T(n) \in \Theta(n^d \log n)$ if $a = b^d$

$T(n) \in \Theta(n^{\log_b a})$ if $a > b^d$